

Comparative Effect of LSVT BIG Protocol versus Nordic Walking on Balance, Gait and Fall Risk in Parkinson's Disease Patients

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Abstract— Parkinson's Disease (PD) is a progressive neurodegenerative disorder marked by bradykinesia, rigidity, postural instability and gait disturbances, leading to increased fall risk and reduced functional independence. LSVT BIG is an intensive amplitude-based exercise protocol designed to recalibrate hypometric movement patterns, whereas Nordic Walking is a task-specific locomotor intervention that enhances gait through external cueing and upper-limb engagement. This randomized controlled trial compared the effects of LSVT BIG and Nordic Walking on balance, gait and fall risk in PD patients. Thirty individuals with idiopathic PD (Hoehn and Yahr Stage II–III) were randomized to Group A (LSVT BIG, n=15) or Group B (Nordic Walking, n=15) for an eight-week supervised program. Outcome measures included Berg Balance Scale (BBS), Timed Up and Go (TUG), GAITRite gait parameters, Falls Efficacy Scale–International (FES-I) and UPDRS-III motor scores, assessed pre- and post-intervention. Both groups showed significant improvements across all measures ($p < 0.001$). LSVT BIG produced greater gains in balance, fall risk reduction and motor function (BBS, TUG, FES-I, UPDRS-III), while Nordic Walking showed relatively larger improvements in gait velocity and cadence. These findings support the use of LSVT BIG as a primary intervention for balance and motor control, with Nordic Walking as a complementary strategy for gait enhancement in PD rehabilitation.

Index Terms— Parkinson's disease, LSVT BIG, Nordic Walking, balance, gait, fall risk, Berg Balance Scale, UPDRS-III.

I. INTRODUCTION

Parkinson's Disease is the second most common neurodegenerative disorder worldwide, characterized by progressive dopaminergic neuron loss in the substantia nigra and consequent basal ganglia circuit dysfunction. The resulting bradykinesia, rigidity, resting tremor and postural instability profoundly impair balance, gait and functional mobility, leading to a high incidence of falls and related injuries.

In addition to reduced gait speed and step length, PD is associated with an amplitude calibration deficit in which patients generate movements that are systematically smaller than intended yet perceive them as normal. This contributes to hypometric postural responses and shuffling gait, increasing fall risk during transfers and ambulation.

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LSVT BIG is an intensive, one-to-one, amplitude-focused treatment that trains patients to perform exaggerated, high-amplitude movements with the cue 'Think BIG, Move BIG.' Repetitive practice of maximum amplitude movements aims to recalibrate the internal perception of movement size and restore more normal scaling in everyday activities.

Nordic Walking, by contrast, uses bilateral poles to provide mechanical support, external rhythmic cueing and upper-limb engagement during walking. It has been shown to improve gait velocity, stride length and endurance in PD by bypassing deficient internal cueing mechanisms and promoting symmetrical, coordinated gait. This study compared these two approaches on balance, gait and fall-related outcomes in Indian PD patients.

II. METHODOLOGY

A. Study Design and Participants

A randomized controlled trial with two parallel groups was conducted in a movement disorders physiotherapy clinic. Thirty patients with idiopathic PD, Hoehn and Yahr Stage II–III, aged 50–75 years, on stable dopaminergic medication and able to walk independently for at least 10 meters were included. Exclusion criteria comprised severe cognitive impairment, other neurological or musculoskeletal disorders affecting gait and recent orthopaedic surgery. Participants were randomly assigned to Group A (LSVT BIG) or Group B (Nordic Walking) using computer-generated randomization.

B. Outcome Measures

Primary outcomes were balance and fall risk assessed by the Berg Balance Scale and Timed Up and Go test. Gait parameters (velocity, cadence, step length) were obtained using a GAITRite walkway system. Fear of falling was measured with the Falls Efficacy Scale–International, and overall motor severity was assessed with the UPDRS-III motor subscale. All assessments were performed in the ON-medication state at baseline and after eight weeks of intervention.

C. Intervention Protocol—LSVT BIG (Group A)

Group A received an expanded LSVT BIG program over eight weeks, consisting of 28 one-hour individual sessions (four sessions per week for four weeks, then three sessions per week for four weeks). Each session included a standardized set of seven maximal-amplitude whole-body exercises, high-amplitude walking tasks, functional component tasks (e.g., sit-to-stand, turning, reaching) and carryover activities. All movements were performed with maximal amplitude under continuous verbal and visual feedback, with daily home exercise assigned on non-clinic days.

D. Intervention Protocol—Nordic Walking (Group B)

Group B participated in an eight-week Nordic Walking program with three to four supervised sessions per week, matched in total training time to the LSVT BIG group. Sessions began with instruction in correct pole technique, followed by progressive outdoor or indoor walking at a moderate intensity. Progression included increasing walking distance and speed, incorporating turns and obstacle negotiation and adding dual-task elements. Group-based sessions with 3–5 participants were used where feasible.

E. Statistical Analysis

Within-group changes were analyzed using paired t-tests and between-group differences in post-intervention outcomes were assessed with independent-samples t-tests. Effect sizes were calculated using Cohen's d, and significance was set at $p < 0.05$.

III. RESULTS

All thirty participants completed the intervention with no serious adverse events such as falls with injury or cardiovascular complications. Baseline demographic and clinical characteristics, including age, sex distribution, disease duration, Hoehn and Yahr stage and initial BBS, TUG, gait, FES-I and UPDRS-III scores, were comparable between groups, indicating successful randomization.

Both groups demonstrated significant within-group improvements in balance, mobility, gait parameters, fall confidence and motor severity ($p < 0.001$). LSVT BIG improved BBS scores from approximately 38.4 to 48.2 points and reduced TUG times from around 17.6 to 13.2 seconds, exceeding minimal clinically important differences and moving most participants above established fall-risk thresholds. Nordic Walking also improved BBS and TUG, but to a lesser extent.

GAITrite analysis revealed that both interventions increased gait velocity, cadence and step length, with Nordic Walking showing slightly greater gains in velocity and cadence, while LSVT BIG produced larger improvements in step length. FES-I scores decreased more in the LSVT BIG group, indicating greater reductions in fear of falling. UPDRS-III motor scores improved significantly in both groups, with a larger mean reduction in the LSVT BIG group. Between-group comparisons showed statistically significant advantages for LSVT BIG on BBS, TUG, FES-I and UPDRS-III and for Nordic Walking on gait velocity and cadence, all with large to very large effect sizes.

IV. DISCUSSION

The present study confirms that both LSVT BIG and Nordic Walking are effective rehabilitation strategies for improving

balance, gait and fall-related outcomes in individuals with mild to moderate Parkinson's Disease. However, the pattern of changes reflects their distinct mechanistic emphases. LSVT BIG, by directly addressing the amplitude calibration deficit through maximal movement training, produced greater improvements in balance, functional mobility and global motor scores.

Nordic Walking, in contrast, capitalized on external rhythmic cueing and upper-limb engagement to enhance gait velocity and cadence, supporting its role as a task-specific locomotor training modality. The poles provided additional base-of-support and stability, which may have facilitated faster and more confident walking without compromising safety.

The greater reduction in FES-I scores observed with LSVT BIG suggests that intensive amplitude training not only improves objective balance performance but also enhances perceived stability and confidence, which are key determinants of activity participation. Given that both interventions exceeded clinically meaningful thresholds on several measures, clinical decision-making can be individualized: LSVT BIG prioritized when balance and overall motor control are the main concerns, and Nordic Walking added or emphasized when gait speed and endurance are primary goals.

Limitations of this study include the relatively small sample size, short-term follow-up and focus on Hoehn and Yahr Stage II–III patients. Future research should investigate combined or sequential LSVT BIG and Nordic Walking protocols, long-term retention of gains and cost-effectiveness in different service delivery models (individual versus group-based training).

V. CONCLUSION

Eight weeks of LSVT BIG or Nordic Walking led to significant improvements in balance, gait and fall-related outcomes in Parkinson's Disease patients. LSVT BIG demonstrated superior effects on balance, functional mobility, fall confidence and overall motor function, while Nordic Walking produced relatively greater gains in gait velocity and cadence. These results support LSVT BIG as a primary intervention for PD motor rehabilitation, with Nordic Walking serving as a valuable adjunct to optimize gait performance and endurance.

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Tables

Variable	LSVT BIG Group		Nordic Walking Group		Notes
Age (years, mean $\pm$ SD)	similar		similar		No significant difference
Sex distribution	similar		similar		Balanced
Disease duration (years)	similar		similar		H&Y II–III
Baseline BBS/TUG/gait/FES-I/UPDRS-III	similar		similar		All $p > 0.05$

Outcome	LSVT BIG Post	Nordic Walking Post	p-value	Effect size	Interpretation
BBS (points)	greater increase	increase	<0.01	Very large	LSVT BIG better balance
TUG (seconds)	larger decrease	decrease	<0.05	Very large	LSVT BIG better fall risk

Gait velocity (m/s)	improved	greater improvement	<0.05	Large	Nordic Walking faster gait
Cadence (steps/min)	improved	greater improvement	<0.05	Large	Nordic Walking more cadence
FES-I (points)	larger reduction	reduction	<0.05	Very large	LSVT BIG better confidence
UPDRS-III (points)	larger reduction	reduction	<0.05	Very large	LSVT BIG better motor control

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